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1. Introduction

This report documents the development of a comprehensive Power BI analytical solution, covering the full process from raw data preparation to the generation of actionable business insights. The primary objective of this project is to build a reproducible and well-structured analytical model that supports real-world decision-making in the banking and financial services sector. The analysis is based on a large-scale loan dataset containing over 45,000 records of loan applicants. The dataset provides detailed information on borrower demographics, financial status, credit history, and loan characteristics, making it suitable for advanced business intelligence analysis.

Throughout this project, several key analytical processes are implemented:

- **Data Cleaning and Transformation:** Addressing data quality issues such as missing values, duplicate records, and inconsistent formats to ensure accuracy and reliability.
- **Data Modelling:** Designing and implementing a relational star schema to support efficient querying, improved performance, and scalable reporting.
- **Analytical Computing:** Developing advanced DAX measures, including performance indicators and risk-related metrics, to enable deeper insights into loan behavior.
- **Data Visualization:** Creating an interactive three-page Power BI dashboard that supports both executive-level summaries and detailed analytical exploration.

Overall, this project demonstrates how raw financial data can be transformed into meaningful insights that support better decision-making in loan approval, risk management, and customer profiling.

2. Data Acquisition and Understanding

2.1 Domain and Business Problem Statement

This project focuses on Banking and Financial Analytics, specifically in the area of loan risk assessment and customer profiling. Financial institutions rely on data-driven insights to evaluate borrowers, manage credit risk, and improve lending decisions.

The main business problem addressed in this project is how to analyze borrower characteristics and loan attributes in order to identify patterns that influence loan approval and default risk. Understanding these patterns is critical for minimizing financial loans and optimizing lending strategies.

Key analytical questions addressed include:

- Which borrower characteristics are associated with higher default risk?
- How do income levels and loan amounts affect loan approval outcomes?
- What role does credit history play in determining loan status?
- Which segments of customers present the highest risk to lenders?
- How can financial institutions improve their loan approval strategies?

By answering these questions, the Power BI solution provides valuable insights that support better credit risk management and more informed decision-making.

2.2 Dataset Source and Overview

The dataset used in this project is a publicly available loan dataset obtained from an online data repository. It contains over 45,000 records of loan applicants and includes both demographic and financial attributes(*Loan Approval Classification Dataset*, n.d.).

Each record represents an individual loan application and includes variables such as age, gender, income, education, employment experience, loan amount, interest rate, credit history length and loan status.

The dataset reflects real-world financial data and includes a mixture of numerical and categorical variable. It also contains some inconsistencies and missing values, making it suitable for demonstrating data cleaning and transformation techniques in Power BI. The dataset is appropriate for business intelligence analysis as it enables the study of borrower behavior, credit risk, and lending performance.

2.3 Description of Key Tables and Fields

The dataset was initially provided as a single flat table containing loan application records. To support efficient analysis and reporting in Power BI, the data was transformed into a structured relational model consisting of fact and dimension tables.

The key tables used in this project are summarized below:

Table Name	Type	Description
Fact_Loans	Fact Table	It contains core loan-related measures such as loan amount, interest rate, and loan status. It serves as the central table for analysis.
Dim_LoanDetails	Dimension	It includes attributes such as loan intent and loan status for categorizing loan types.
Dim_Credit	Dimension	It contains credit-related attributes such as credit score and credit history length.
Dim_HomeOwnership	Dimension	It stores information about the borrower's home ownership status.

2.4 Sustainability for Advanced Power BI Analysis

The selected dataset is highly suitable for advanced Power BI analysis for several reasons:

- **Volume of the dataset:** This dataset contains a large volume of records which exceeds the minimum requirements and allows for meaningful aggregation and pattern detection.
- **Variety of data types:** The dataset includes both numerical and categorical variables. Numerical fields such as income, loan amount and interest rate enable quantitative analysis, while categorical variables such as education level, loan intent, and home ownership allow for segmentation and comparison.
- **Data Transformation:** The dataset contains several data quality issues. These issues provide an opportunity to demonstrate data cleaning and transformation skills using Power Query.
- **Advanced Analytical Techniques:** The dataset enables the application of advanced analytical techniques such as DAX calculations, segmentation and trend analysis, making it ideal for building a comprehensive and professional Power BI solution.

Evidence for this section

The screenshot will show the first few rows of the dataset:

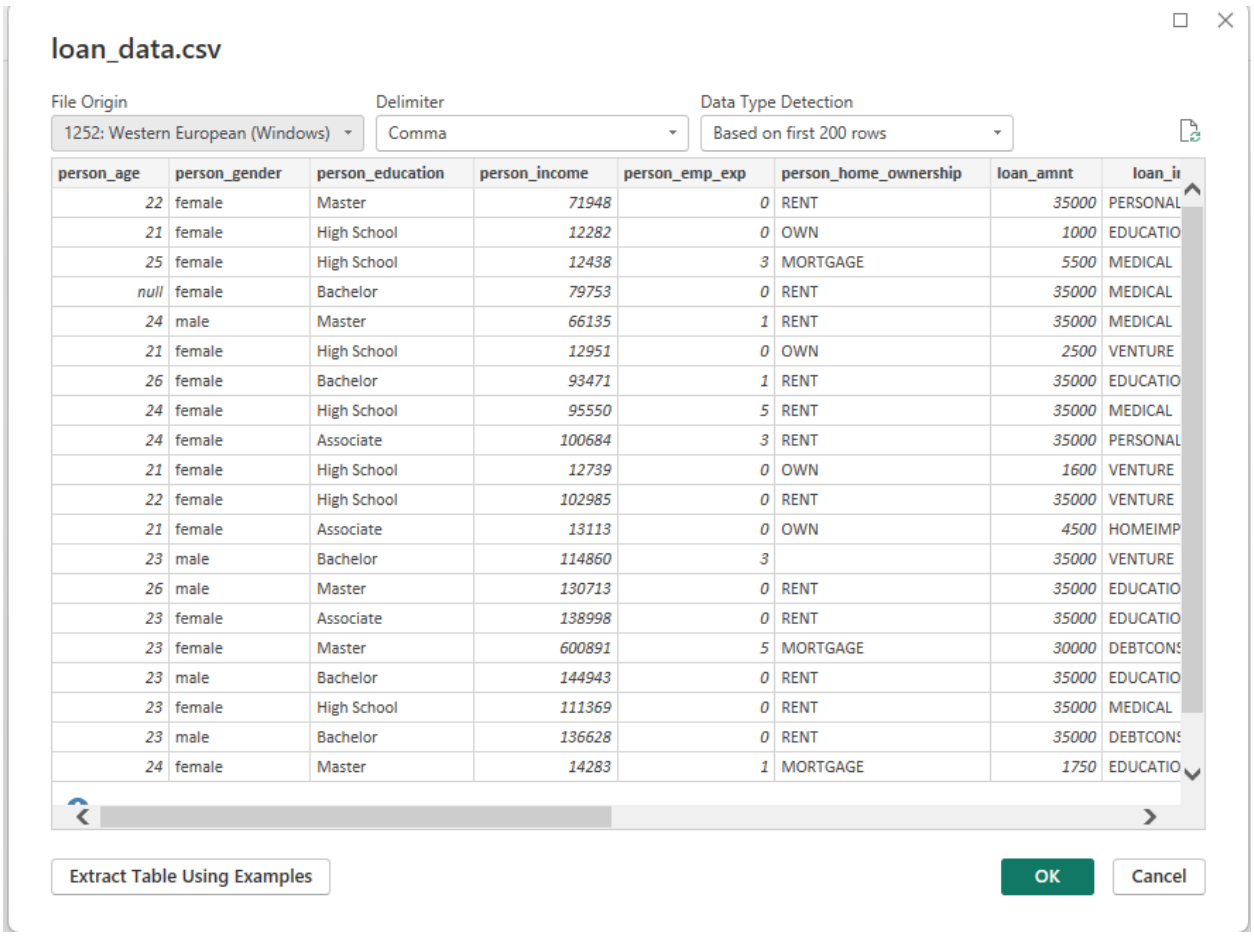
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	person_a	person_ge	person_ec	person_in	person_er	person_h	loan_amn	loan_inter	loan_int_r	loan_perc	cb_persor	credit_scc	previous_l	loan_status
2	22	female	Master	71948	0	RENT	35000	PERSONAI	16.02	0.49	3	561	No	1
3	21	female	High Scho	12282	0	OWN	1000	EDUCATIO	11.14	0.08	2	504	Yes	0
4	25	female	High Scho	12438	3	MORTGAG	5500	MEDICAL	12.87	0.44	3	635	No	1
5		female	Bachelor	79753	0	RENT	35000	MEDICAL	15.23	0.44	2	675	No	1
6	24	male	Master	66135	1	RENT	35000	MEDICAL	14.27	0.53	4	586	No	1
7	21	female	High Scho	12951	0	OWN	2500	VENTURE	7.14	0.19	2	532	No	1
8	26	female	Bachelor	93471	1	RENT	35000	EDUCATIO	12.42	0.37	3	701	No	1
9	24	female	High Scho	95550	5	RENT	35000	MEDICAL	11.11	0.37	4	585	No	1
10	24	female	Associate	100684	3	RENT	35000	PERSONAI	8.9	0.35	2	544	No	1
11	21	female	High Scho	12739	0	OWN	1600	VENTURE	14.74	0.13	3	640	No	1
12	22	female	High Scho	102985	0	RENT	35000	VENTURE	10.37	0.34	4	621	No	1
13	21	female	Associate	13113	0	OWN	4500	HOMEIMP	8.63	0.34	2	651	No	1
14	23	male	Bachelor	114860	3		35000	VENTURE	7.9	0.3	2	573	No	1
15	26	male	Master	130713	0	RENT	35000	EDUCATIO	18.39	0.27	4	708	No	1
16	23	female	Associate	138998	0	RENT	35000	EDUCATIO	7.9	0.25	4		No	0
17	23	female	Master	600891	5	MORTGAG	30000	DEBTCON:	10.65	0.05	3	670	Yes	0
18	23	male	Bachelor	144943	0	RENT	35000	EDUCATIO	7.9		4	663	No	0
19	23	female	High Scho	111369	0	RENT	35000	MEDICAL	20	0.31	4	694	No	1
20	23	male	Bachelor	136628	0	RENT	35000	DEBTCON:	18.25	0.26	4	709	No	1
21	24	female	Master	14283	1	MORTGAG	1750	EDUCATIO	10.99	0.12	2	679	No	1
22	25	male	Bachelor	195718	0	RENT	35000	VENTURE	7.49	0.18	4	684	Yes	0
23	25	male	High Scho	165792	4	RENT	34800	PERSONAI	16.77	0.21	2	662	No	0
24	22	female	Master	79255	0	RENT	34000	EDUCATIO	17.58	0.43	4	691	No	1
25	24	female	Bachelor	13866	0	OWN	1500	PERSONAI	7.29	0.11	3	600	Yes	0
26	22	male		97420	1	RENT	33950	PERSONAI	14.54	0.35	4	691	No	1

loan_data

3. Data Cleaning and Transformation in Power Query

3.1 Data Import

The dataset was loaded into Power BI using the “Get Data” feature. Upon loading into Power Query, the dataset was examined to understand its structure, identify data types, and detect potential issues such as missing values, duplicates, and inconsistencies.



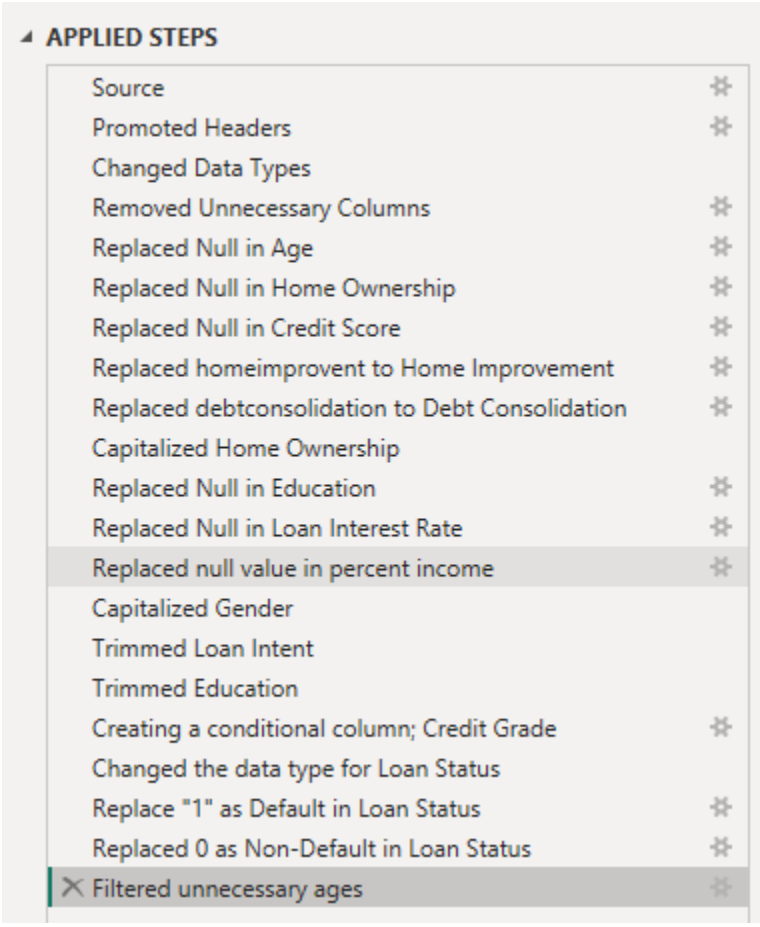
person_age	person_gender	person_education	person_income	person_emp_exp	person_home_ownership	loan_amnt	loan_in
22	female	Master	71948	0	RENT	35000	PERSONAL
21	female	High School	12282	0	OWN	1000	EDUCATIO
25	female	High School	12438	3	MORTGAGE	5500	MEDICAL
null	female	Bachelor	79753	0	RENT	35000	MEDICAL
24	male	Master	66135	1	RENT	35000	MEDICAL
21	female	High School	12951	0	OWN	2500	VENTURE
26	female	Bachelor	93471	1	RENT	35000	EDUCATIO
24	female	High School	95550	5	RENT	35000	MEDICAL
24	female	Associate	100684	3	RENT	35000	PERSONAL
21	female	High School	12739	0	OWN	1600	VENTURE
22	female	High School	102985	0	RENT	35000	VENTURE
21	female	Associate	13113	0	OWN	4500	HOMEIMP
23	male	Bachelor	114860	3		35000	VENTURE
26	male	Master	130713	0	RENT	35000	EDUCATIO
23	female	Associate	138998	0	RENT	35000	EDUCATIO
23	female	Master	600891	5	MORTGAGE	30000	DEBTCONS
23	male	Bachelor	144943	0	RENT	35000	EDUCATIO
23	female	High School	111369	0	RENT	35000	MEDICAL
23	male	Bachelor	136628	0	RENT	35000	DEBTCONS
24	female	Master	14283	1	MORTGAGE	1750	EDUCATIO

3.2 Data Cleaning and Transformation

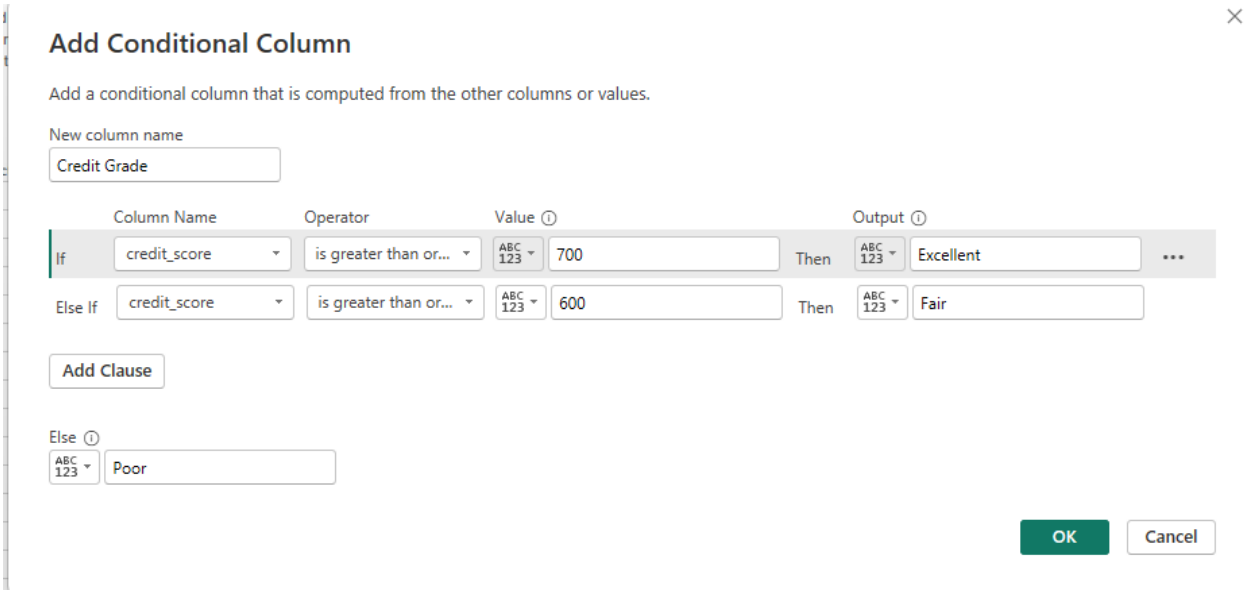
To ensure the loan dataset was accurate for financial risk modeling, I performed the following transformations:

- Identified null values within the loan_int_rate column. I utilized the "Replace Values" transformation to fill these gaps, ensuring that average interest rate calculations remain mathematically sound.
- Removed currency symbols and percentage characters from the raw text fields. This was a prerequisite for converting these columns into formats suitable for mathematical aggregation.
- Transformed the loan_status column from 0/1 to descriptive terms ("Non-Default" / "Default") to make the final dashboard intuitive for non-technical stakeholders.
- Changed the case for person_gender, loan_intent and person_home_ownership
- Explicitly changed data types for financial metrics to support complex DAX measures.
- Applied Trim and Clean functions to text-based columns such as person_education and loan_intent. This removed trailing spaces and non-printable characters that often cause errors in filtering and slicers.
- I added a new column named **"Credit Grade"** using IF/THEN logic:
    - If credit_score >= 700, then **"Excellent"**
    - If credit_score >= 600, then **"Fair"**
    - Else, **"Poor"**.
- Filtered the person_age and person_income columns to remove unrealistic values (e.g., ages over 100 or negative income) that would skew the risk profile.

Applied Steps



Conditional Column



4. Data Modelling

4.1 Star Schema Architecture

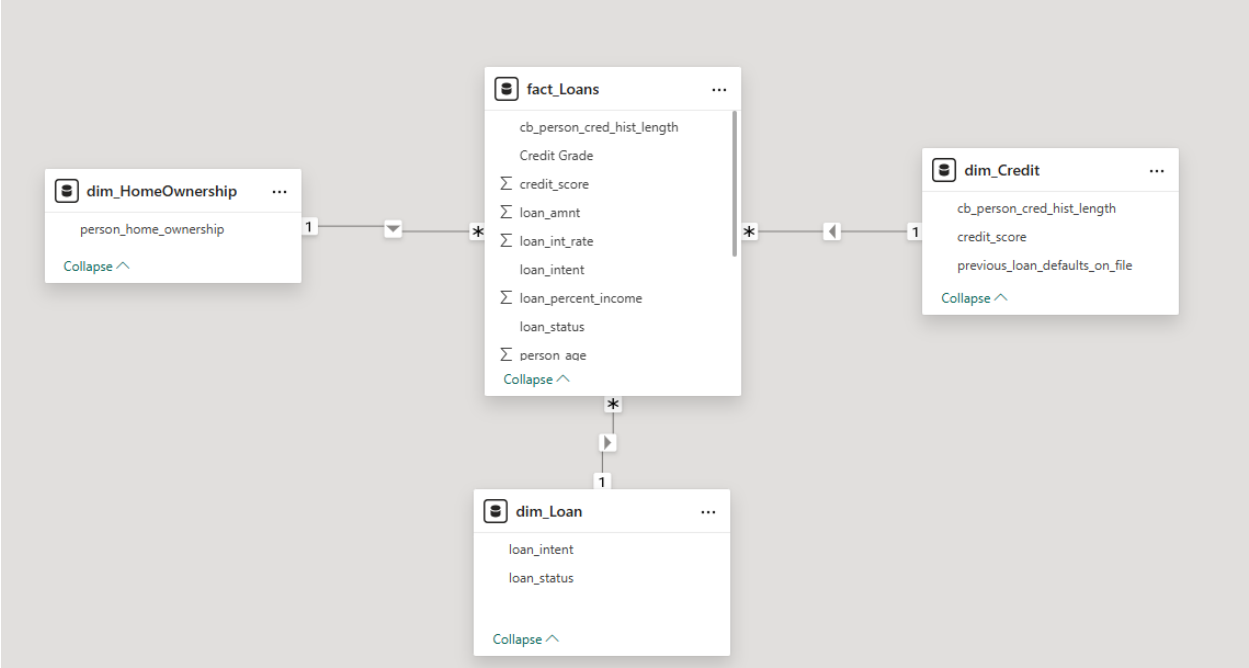
The original dataset was provided as a single flat table containing lan application records, without a predefined relational structure. To support efficient and analytical performance in Power BI, the dataset was transformed into a structured data model.

A star schema approach was adopted, where a central fact table stores measurable loan-related data, and surrounding dimension tables provide descriptive attributes for filtering and segmentation. This structure improves query performance, enhances scalability, and simplifies report development.

During the modeling process, the dimension tables were created. Several dimension tables were created to support filtering and categorization:

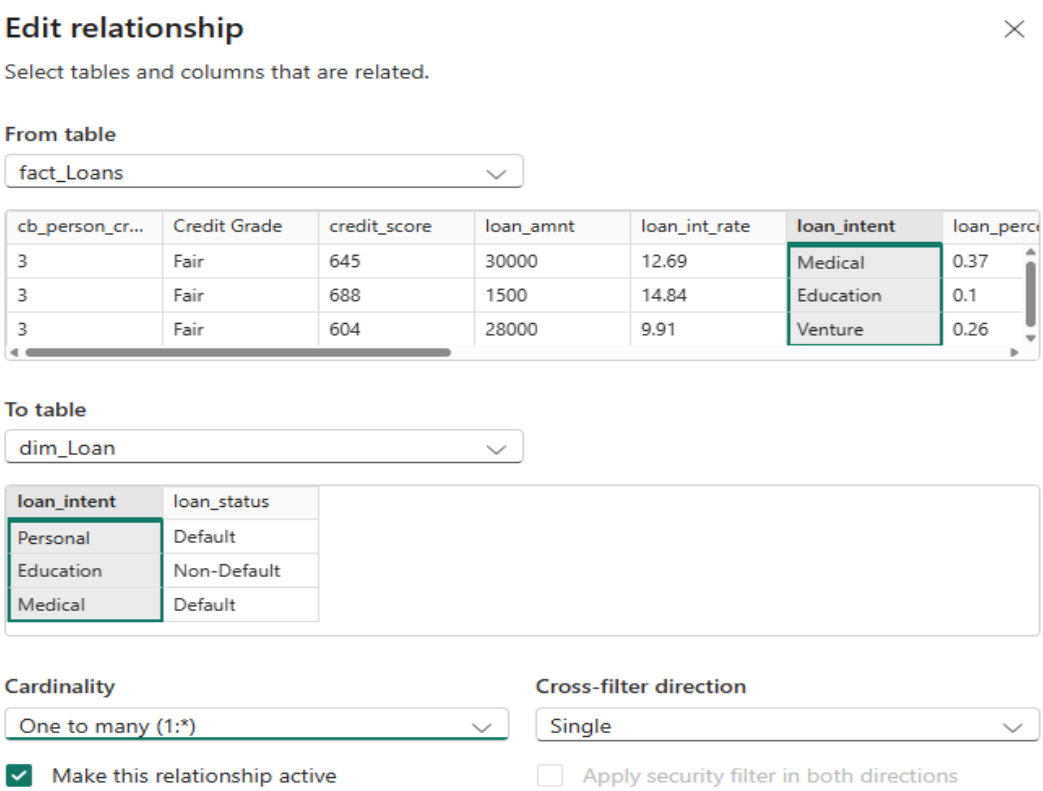
- **Dim_Loan** – contains loan-related categories such as loan intent and loan status.
- **Dim_HomeOwnership** – contains borrower home ownership categories.
- **Dim_Credit** – contains credit-related attributes such as credit score and credit history length.

An attempt was made to create a borrower dimension table; however, due to the absence of a unique borrower identifier, it was not possible to generate a meaningful and non-redundant dimension. As a result, borrower attributes were retained within the fact table to preserve data integrity. This modeling approach ensures a balance between normalization and analytical usability.


fact_Loans
cb_person_cred_hist_length
Credit Grade
credit_score
loan_amnt
loan_int_rate
loan_intent
loan_percent_income
loan_status
person_age
Collapse
dim_HomeOwnership
person_home_ownership
Collapse
dim_Credit
cb_person_cred_hist_length
credit_score
previous_loan_defaults_on_file
Collapse
dim_Loan
loan_intent
loan_status
Collapse
1
*
*
1
*
1

4.2 Relationship Management and Cardinality

As shown in the model view, I established **One-to-many(1:*)** relationships between these dimensions and the fact table. This ensures that descriptive attributes such as education level, gender, or loan purpose can effectively filter the transactional data without creating data redundancy. To maintain strict data integrity and prevent ambiguity in the results, the cross-filter direction for these relationships was set to **"Single",** ensuring that filters flow logically from the descriptive dimensions to the fact table.



4.3 Time Intelligence and Optimization

A dedicated date table was created to support time-based analysis and enable the use of time intelligence functions in Power BI. The date table includes fields such as year, month, quarter, and day , and was generated using DAX. The table allows for consistent time-based filtering and supports calculations such as trends and period comparisons

Although the original dataset does not contain a strong temporal component, the inclusion of a date table enhances the analytical capabilities of the model and aligns with the best practices in Power BI development.

```
Date_Table =
ADDCOLUMNS(
    CALENDAR(DATE(2015,1,1), DATE(2025,12,31)),
    "Year", YEAR([Date]),
    "Month", FORMAT([Date], "MMMM"),
    "Month Number", MONTH([Date]),
    "Quarter", "Q" & FORMAT([Date], "Q")
)
```

The first few rows found in the **Date Table**:

Date	Year	Month	Month Number	Quarter
01/07/2015 12:00:00 AM	2015	July	7	Q3
02/07/2015 12:00:00 AM	2015	July	7	Q3
03/07/2015 12:00:00 AM	2015	July	7	Q3
04/07/2015 12:00:00 AM	2015	July	7	Q3
05/07/2015 12:00:00 AM	2015	July	7	Q3
06/07/2015 12:00:00 AM	2015	July	7	Q3
07/07/2015 12:00:00 AM	2015	July	7	Q3
08/07/2015 12:00:00 AM	2015	July	7	Q3
09/07/2015 12:00:00 AM	2015	July	7	Q3
10/07/2015 12:00:00 AM	2015	July	7	Q3
11/07/2015 12:00:00 AM	2015	July	7	Q3
12/07/2015 12:00:00 AM	2015	July	7	Q3
13/07/2015 12:00:00 AM	2015	July	7	Q3
14/07/2015 12:00:00 AM	2015	July	7	Q3
15/07/2015 12:00:00 AM	2015	July	7	Q3
16/07/2015 12:00:00 AM	2015	July	7	Q3
17/07/2015 12:00:00 AM	2015	July	7	Q3
18/07/2015 12:00:00 AM	2015	July	7	Q3
19/07/2015 12:00:00 AM	2015	July	7	Q3
20/07/2015 12:00:00 AM	2015	July	7	Q3
21/07/2015 12:00:00 AM	2015	July	7	Q3
22/07/2015 12:00:00 AM	2015	July	7	Q3
23/07/2015 12:00:00 AM	2015	July	7	Q3

5. DAX Measures and Calculated Columns

5.1 DAX Measures

5.1.1 Total Loan Amount

```
Total Loan Amount = SUM(fact_Loans[loan_amnt])
```

This measure calculates the total value of all loans issued. It helps assess the overall lending volume of the institution.

5.1.2 Average Loan Amount

```
Average Loan Amount = AVERAGE(fact_Loans[loan_amnt])
```

This measure calculates the average loan size, providing insight into typical borrowing behavior.

5.1.3 Total Borrowers

```
Total Borrowers = COUNTROWS(fact_Loans)
```

This measure counts the total number of loan applications, representing the number of borrowers analyzed.

5.1.4 Average Interest Rate

```
Average Interest Rate = AVERAGE(fact_Loans[loan_int_rate])
```

This measure calculates the average interest rate applied across all loans, helping evaluate lending conditions.

5.1.5 Default Rate (%)

```
Default Rate % =
DIVIDE(
    CALCULATE(COUNTROWS(fact_Loans), fact_Loans[loan_status] = "Default"),
    COUNTROWS(fact_Loans)
)
```

This measure calculates the proportion of loans that have defaulted, providing a key indicator of credit risk.

5.1.6 Approved Loan Rate (%)

```
Approved Loan Rate % =
DIVIDE(
    CALCULATE(COUNTROWS(fact_Loans), fact_Loans[loan_status] = "Approved"),
    COUNTROWS(fact_Loans)
)
```

This measure shows the percentage of approved loans, helping evaluate lending success rates.

5.1.7 Average Loan-to-Income Ratio

```
Avg Loan % Income = AVERAGE(fact_Loans[loan_percent_income])
```

This measure evaluates how much of a borrower's income is allocated to loan repayment, indicating financial burden.

5.1.8 High-Risk Loans Count

```
High Risk Loans =
CALCULATE(
    COUNTROWS(fact_Loans),
    fact_Loans[loan_percent_income] > 0.4
)
```

This measure identifies loans where repayment exceeds 40% of income, indicating high financial risk.

5.1.9 Loan Ranking by Amount

```
Loan Rank =
RANKX(
    ALL(fact_Loans),
    [loan_amnt],
    ,
    DESC
)
```

This measure ranks loan values to identify the largest loans issued.

5.2 Calculate Columns

5.2.1 Income Band

```
Income Band =
SWITCH(
    TRUE(),
    fact_Loans[person_income] < 30000, "Low Income",
    fact_Loans[person_income] < 70000, "Medium Income",
    "High Income"
)
```

This column categorizes borrowers into income groups to support segmentation analysis.

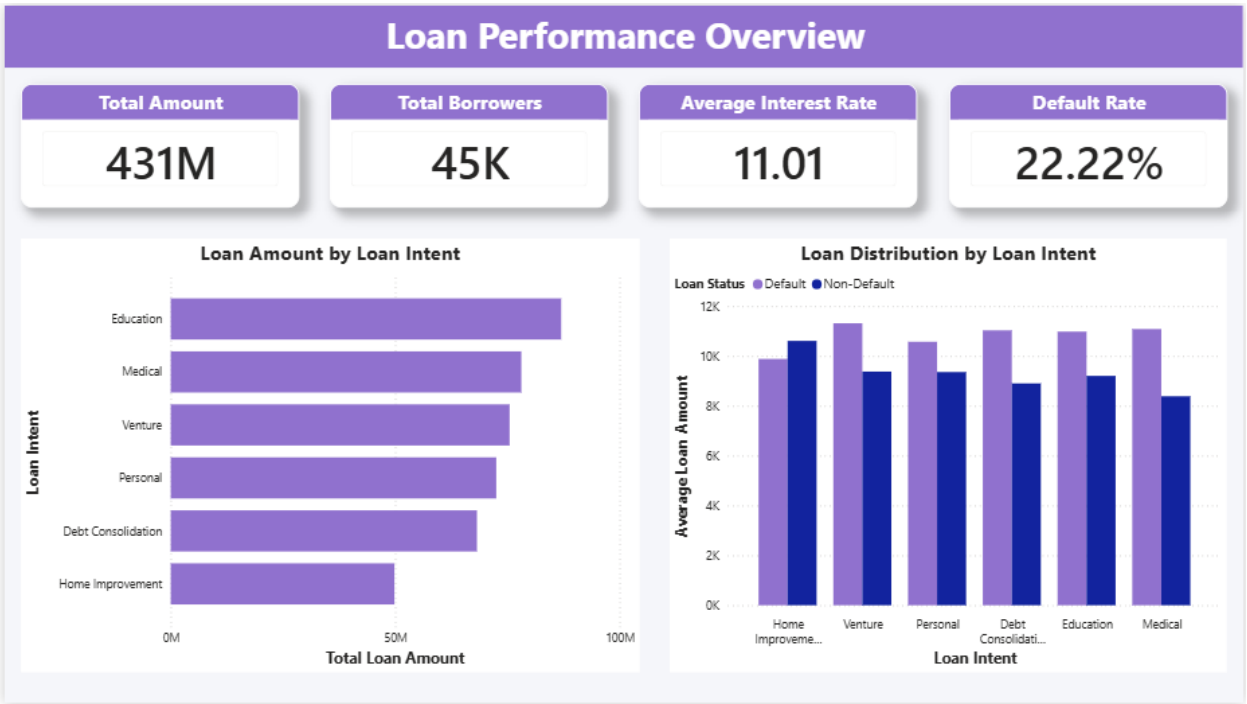
5.2.2 Risk Category

```
Risk Category =
IF(
    fact_Loans[loan_percent_income] > 0.4,
    "High Risk",
    "Low Risk"
)
```

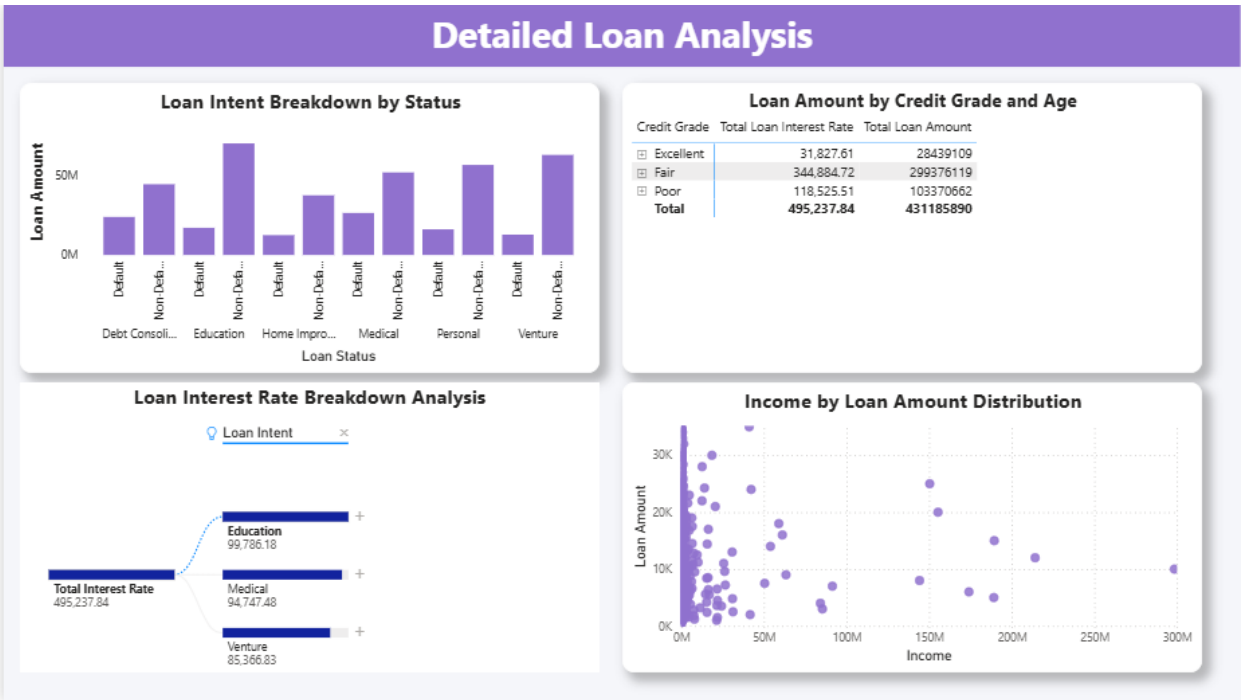
This column classifies loans based on repayment burden, helping identify high-risk borrowers.

6. Dashboard Design and Visualization

Page 1: **Loan Performance Overview**



Page 2: **Detailed Loan Analysis**



7. Analytical Insights and Business Recommendations

7.1 Analytical Insights

Based on the analysis conducted using the Power BI dashboard, several key insights were identified from the loan dataset.

7.1.1 High Loan Burden Increases Default Risk

It was observed that borrowers with a high loan-to-income ratio (loan_percent_income) are significantly more likely to default on their loans. Loans exceedingly approximately 40% of a borrower's income show a noticeable increase in default rates, indicating financial strain.

7.1.2 Lower Income Borrowers Exhibit Higher Risk

Borrowers classified under the low-income band tend to have a higher proportion of loan defaults compared to medium- and high-income groups. This suggests that income level is a critical factor in assessing repayment capability.

7.1.3 Certain Loan Intent Categories Carry Higher Risk

Analysis of loan intent categories shows that some types of loans, such as personal or high-risk purposes, are more likely to result in defaults compared to more structured or essential loan categories. This indicates that loan purpose influences repayment behavior.

7.1.4 Credit History Plays a Key Role in Loan Outcomes

Borrowers with shorter credit history lengths and lower credit scores tend to have higher default rates. In contrast, individuals with stronger credit profiles are more likely to have approved and successfully repaid loans.

7.1.5 Home Ownership is Associated with Lower Default Rates

Borrowers who own homes generally exhibit lower default rates compared to renters or individuals with less stable housing situations. This suggests that financial stability and asset ownership contribute positively to loan repayment behavior.

7.1.6 Interest Rates Tend to Be Higher for Riskier Borrowers

Higher interest rates are often associated with borrowers who have lower credit scores or higher loan risk indicators. This reflects risk-based pricing strategies used by financial institutions.

7.2 Business Recommendations

Based on the insights derived from the analysis, the following recommendations are proposed to improve loan performance and risk management:

7.2.1 Implement Stricter Controls on High Loan-to-Income Ratios

Financial institutions should introduce stricter approval criteria for borrowers whose loan repayment exceeds a certain percentage of their income (e.g., 40%). This can help reduce the likelihood of default and improve portfolio quality.

7.2.2 Enhance Risk-Based Pricing Strategies

Interest rates should be more closely aligned with borrower risk profiles. Borrowers with lower credit scores or weaker financial positions should be carefully evaluated, and pricing strategies should reflect the associated risk of mitigating potential losses.

7.2.3 Prioritize Lending to Lower-Risk Segments

Lenders should focus on customers with stable income levels, strong credit histories, and home ownership status, as these groups demonstrate lower default rates. Targeting such segments can improve overall loan performance.

7.2.4 Improve Credit Assessment Models

Incorporating additional variables such as credit history length, previous defaults, and income stability into credit scoring models can enhance the accuracy of loan approval decisions.

7.2.5 Offer Financial Education and Support Programs

Providing financial literacy programs for high-risk borrowers may help improve repayment behavior and reduce default rates over time.

GitHub Link: https://github.com/l-kamau/DSA3050-ADVANCED-POWERBI-ANALYSIS